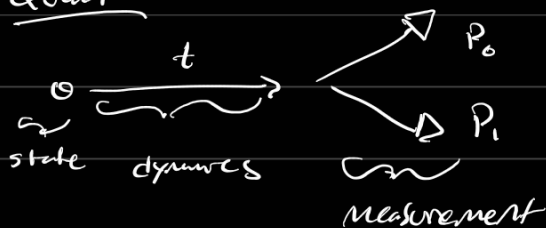
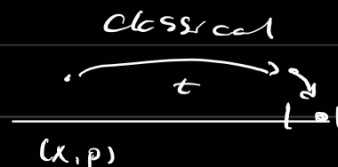


Quant

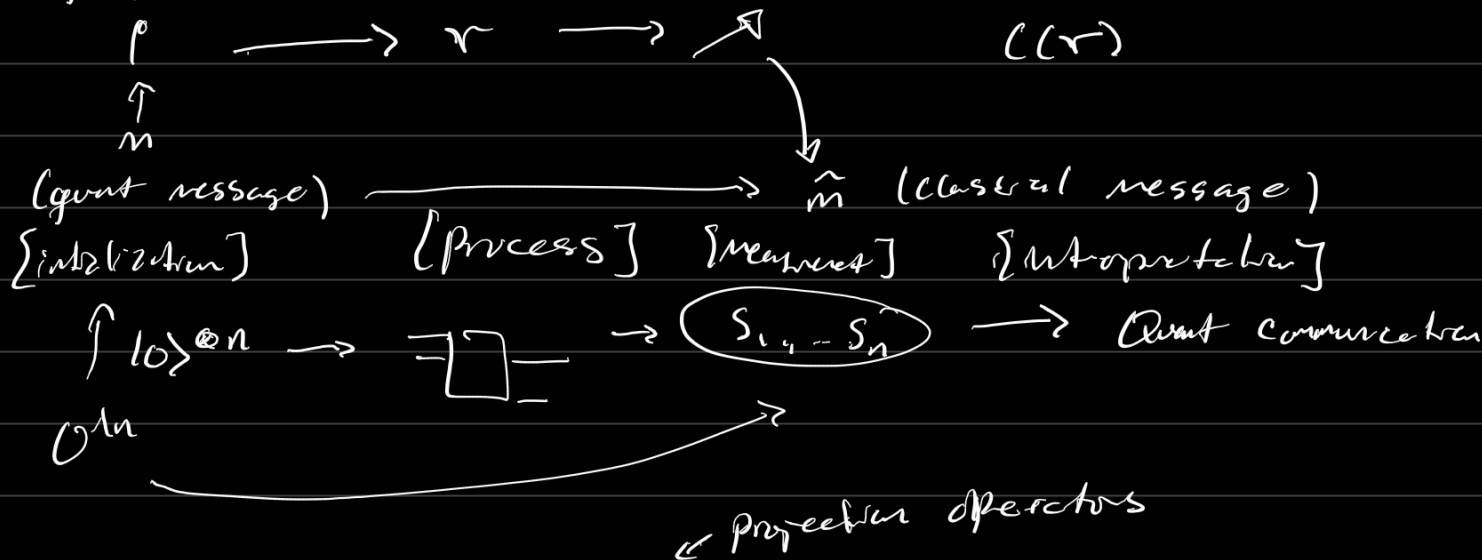


$$\langle f \rangle_p = \sum_n f_n P_n$$



Quantum bit

• preparation transmission detection interpretation



Measurement: POVM $\{M_i\}$ $M_i \geq 0$, $\sum_i M_i = \mathbb{I}$

$$P_i = \langle \psi | M_i | \psi \rangle$$

$$M_0, P_0 = \langle \psi | M_0 | \psi \rangle$$

$$M_1, P_1 = \langle \psi | M_1 | \psi \rangle$$

Observables: (self-adjoint)

$$A = A^\dagger = \sum_i a_i M_i, \quad a_i \in \mathbb{R} \quad \{M_i\}: \text{POVM}$$

$$\langle a \rangle_p = \sum_i a_i P_i = \sum_i a_i \langle \psi | M_i | \psi \rangle = \langle \psi | A | \psi \rangle$$

$$\langle a \rangle_p = \langle A \rangle_\psi$$

Qubits:

$$|H_2\rangle = 2, \quad H_2 = \text{span}\{|0\rangle, |1\rangle\}, \quad \alpha, \beta \in \mathbb{C}$$

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$|0\rangle \langle 0| = M_0$$

$$|1\rangle \langle 1| = M_1$$

$$P_0 = \langle \psi | 0 \rangle \langle 0 | \psi \rangle = |\alpha|^2$$

$$P_1 = \langle \psi | 1 \rangle \langle 1 | \psi \rangle = |\beta|^2$$

$$|\alpha|^2 + |\beta|^2 = 1$$

$$r_1^2 + r_2^2 = 1$$

$$\forall \text{ Unit state: } |\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle$$

Pauli: $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

Hadamard: $H \equiv \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

$\sigma_x |0\rangle = |1\rangle$ — [X] — = — [not] —

$\sigma_x |1\rangle = |0\rangle$ $\{ |0\rangle, |1\rangle \}$: Computational basis

Proof: $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ ✓
 $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ ✓

$H|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \equiv |+\rangle$

$H|1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) \equiv |-\rangle$

$X \equiv \sigma_x$
 $Y \equiv \sigma_y$
 $Z \equiv \sigma_z$

$Z = |0\rangle\langle 0| - |1\rangle\langle 1|$
 $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$

lets note some of this

$\left\{ \begin{array}{l} |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } \langle 0| = \begin{pmatrix} 1 & 0 \end{pmatrix} \\ |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{ and } \langle 1| = \begin{pmatrix} 0 & 1 \end{pmatrix} \\ |0\rangle\langle 0| = (10) \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \checkmark \\ |1\rangle\langle 1| = (01) \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \checkmark \end{array} \right\}$

thus $|+\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$
 $|-\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$

Note: $X = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = |1\rangle\langle 0| + |0\rangle\langle 1|$
 $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$

$(10) \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ $(01) \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$

Claim: $X = H Z H^\dagger$ so $\sigma_x \sim \sigma_z$

and $X = |+\rangle\langle +| - |-\rangle\langle -|$

proof: $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = |1\rangle\langle 0| + |0\rangle\langle 1|$

$|+\rangle\langle +| = \frac{1}{2} (|0\rangle + |1\rangle)(\langle 0| + \langle 1|)$

$= \frac{1}{2} (|0\rangle\langle 0| + |1\rangle\langle 0| + |0\rangle\langle 1| + |1\rangle\langle 1|)$

$= \frac{1}{2} (|0\rangle\langle 0| + X + |1\rangle\langle 1|)$

$$|-\rangle\langle-| = \frac{1}{2}(|0\rangle - |1\rangle)(\langle 0| - \langle 1|)$$

$$= \frac{1}{2}(|0\rangle\langle 0| - |0\rangle\langle 1| - |1\rangle\langle 0| + |1\rangle\langle 1|)$$

$$|+\rangle\langle+| - |-\rangle\langle-| = X \quad \checkmark$$

$$Z = |0\rangle\langle 0| - |1\rangle\langle 1|$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$H|0\rangle = |+\rangle, \quad H|1\rangle = |-\rangle$$

$$HZ = (H|0\rangle)\langle 0| - (H|1\rangle)\langle 1|$$

$$= |+\rangle\langle 0| - |-\rangle\langle 1|$$

$$HZH^\dagger = |+\rangle\langle+| - |-\rangle\langle-| = X$$

$$H^2 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = I \quad \checkmark$$

$$HXH^\dagger = Z$$

* Phase operation: $P(\alpha) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\alpha} \end{pmatrix}$ on

$$|\psi(\alpha, \phi)\rangle = \cos(\frac{\phi}{2})|0\rangle + e^{i\phi} \sin(\frac{\phi}{2})|1\rangle$$

$$|\psi_0\rangle \xrightarrow{U_t} |\psi_t\rangle \quad i\hbar \frac{d}{dt} |\psi_t\rangle = \hat{H} |\psi_t\rangle$$

$$|\psi_t\rangle = \underbrace{e^{-\frac{i\hat{H}t}{\hbar}}}_{U(t)} |\psi_0\rangle$$

Take Qubit state:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$\begin{aligned} \text{D } M_0 = |0\rangle\langle 0| \quad P_0 &= |\alpha|^2 \\ \text{D } M_1 = |1\rangle\langle 1| \quad P_1 &= |\beta|^2 \end{aligned}$$

Cases: prepare

$$|0\rangle \xrightarrow{H} \begin{cases} \text{D } M_0 = |0\rangle\langle 0| \quad P_0 = \frac{1}{2} \\ \text{D } M_1 = |1\rangle\langle 1| \quad P_0 = \frac{1}{2} \end{cases}$$

$$H|0\rangle = |+\rangle = |\psi\rangle$$

$$|0\rangle \xrightarrow{H} \begin{cases} \text{D } M_0 \quad P_0 = \langle \psi | M_0 | \psi \rangle \\ \text{D } M_1 \quad P_1 = \langle \psi | M_1 | \psi \rangle \end{cases}$$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$|+\rangle \quad \frac{1}{\sqrt{2}}(|0\rangle + e^{i\alpha}|1\rangle) \quad \frac{1}{\sqrt{2}}(|+\rangle + e^{i\alpha}|-\rangle) = |\psi\rangle$$

$$P = \frac{1}{2}(|+\rangle + e^{-i\alpha}|-\rangle)(\langle+| + e^{i\alpha}\langle-|) = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1| + e^{i\alpha}|-\rangle\langle+| + e^{-i\alpha}|+\rangle\langle-|)$$

$$P_0 = \frac{1}{2} (\langle 11 | + e^{-i\alpha} \langle -11 |) (|0\rangle + e^{i\alpha} |1\rangle) \\
= \frac{1}{2} \frac{1}{2} (1 + e^{-i\alpha}) (1 + e^{i\alpha}) \\
= \frac{1}{2} \frac{1}{2} (2 + e^{i\alpha} + e^{-i\alpha})$$

$$P_0 = \frac{1}{2} (1 + \cos(\alpha))$$

$$P_0 + P_1 = 1$$

$$P_1 = 1 - \frac{1}{2} - \frac{1}{2} \cos(\alpha)$$

$$P_1 = \frac{1}{2} - \frac{1}{2} \cos(\alpha)$$

Q3

$$|+\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

$$|+\rangle \xrightarrow{Z} \begin{cases} D M_0 = |0\rangle\langle 0| \\ D M_1 \end{cases}$$

$$Z|+\rangle = |4\rangle = (|0\rangle\langle 0| - |1\rangle\langle 1|) \left(\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \right) \\
= \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) \\
= |-\rangle$$

$$Z|-\rangle = \frac{1}{\sqrt{2}} (|0\rangle\langle 0| - |1\rangle\langle 1|) (|0\rangle - |1\rangle) \\
= |+\rangle$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} b \\ a \end{pmatrix} \quad a = b$$

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \checkmark$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix} =$$

$$P(\phi) |+\rangle = (|0\rangle\langle 0| + e^{i\phi} |1\rangle\langle 1|) \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \\
= \frac{1}{\sqrt{2}} (|0\rangle + e^{i\phi} |1\rangle)$$

$$P_\phi = \langle 4 | + \rangle \langle + | 4 \rangle \\
= \frac{1}{2} (\langle 0 | + e^{-i\phi} \langle 1 |) (|0\rangle + |1\rangle) \\
= \frac{1}{2} (1 + e^{-i\phi}) (1 + e^{i\phi}) \\
= \frac{1}{2} (2 + e^{i\phi} + e^{-i\phi}) \\
= \frac{1}{2} (1 + \cos(\phi)) = 0$$

$$\phi = \pi$$